

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

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Section Id :	640653134544
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	26
Number of Questions to be attempted :	26
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653374404
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532039932 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING

(COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536642026.  Yes

6406536642027.  No

Sub-Section Number :

2

Sub-Section Id :

640653374405

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532039933 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software feature has already been released. Developers observe that users are struggling with the interface on a specific browser and release a patch to fix it.

This activity belongs to which phase?

Options :

6406536642028.  Design

6406536642029.  Development

6406536642030.  Testing

6406536642031.  Maintenance

Question Number : 3 Question Id : 6406532039934 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Two developers are working on different modules of an e-commerce system. One developer modifies a function's parameters, causing the other developer's code to break.

Which concept would have helped prevent this issue?

Options :

6406536642032.  Unit testing

6406536642033.  Interface definition

6406536642034.  Acceptance testing

6406536642035.  Code refactoring

Question Number : 4 Question Id : 6406532039935 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Before releasing a banking application, the development team tests individual functions such as “check balance” and “transfer money” in complete isolation.

What type of testing is being performed?

Options :

6406536642036. ✖ Integration testing

6406536642037. ✖ Acceptance testing

6406536642038. ✔ Unit testing

6406536642039. ✖ Beta testing

Question Number : 5 Question Id : 6406532039936 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A student team begins coding a project immediately after an informal discussion with users. After completion, they repeatedly modify the code to fix mismatches with user expectations.

This approach is best described as:

Options :

6406536642040. ✔ Build-and-fix programming

6406536642041. ✖ Waterfall model

6406536642042. ✖ Spiral model

6406536642043. ✖ Prototype model

Question Number : 6 Question Id : 6406532039938 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software team wants to collect answers to specific, well-defined questions from thousands of geographically distributed sellers.

Which requirement-gathering technique is most appropriate?

Options :

6406536642048. ✖ Interviews

6406536642049. ✖ Focus groups

6406536642050. ✖ Observation

6406536642051. ✔ Questionnaires

Question Number : 7 Question Id : 6406532039939 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following is a correct user story template as per BDD?

Options :

- 6406536642052. ✖ Feature–Role–Time
- 6406536642053. ✖ Input–Process–Output
- 6406536642054. ✔ Role–Feature–Benefit
- 6406536642055. ✖ Task–Module–Deadline

Question Number : 8 Question Id : 6406532039940 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An e-commerce website allows returning customers to purchase items quickly by saving delivery addresses and payment details.

This design primarily improves:

Options :

- 6406536642056. ✖ Effectiveness
- 6406536642057. ✔ Efficiency
- 6406536642058. ✖ Learnability
- 6406536642059. ✖ Safety

Question Number : 9 Question Id : 6406532039941 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Multiple screens of a mobile application are drawn on paper, and switching between screens is simulated manually during discussions with users.

Which prototyping technique is being used?

Options :

- 6406536642060. ✖ Storyboarding
- 6406536642061. ✔ Paper prototyping
- 6406536642062. ✖ Digital mock-ups
- 6406536642063. ✖ Coding a beta version

Question Number : 10 Question Id : 6406532039942 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An e-commerce website allows users to browse products and check prices without requiring them to log in.

Which heuristic for action does this design decision support?

Options :

- 6406536642064. ✔ Freedom

6406536642065. ✖ Flexibility

6406536642066. ✖ Recognition over recall

6406536642067. ✖ Consistency

Question Number : 11 Question Id : 6406532039944 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A project has an initial COCOMO effort estimate of 20 person-months. After considering cost drivers, the Effort Adjustment Factor (EAF) is 1.5.

What is the final estimated effort?

Options :

6406536642072. ✖ 20 PM

6406536642073. ✖ 25 PM

6406536642074. ✔ 30 PM

6406536642075. ✖ 35 PM

Question Number : 12 Question Id : 6406532039945 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A person ensures that the development team delivers maximum value to the client and prioritizes work considering business needs and technical constraints.

Which Scrum role is being described?

Options :

6406536642076. ✖ Scrum Master

6406536642077. ✔ Product Owner

6406536642078. ✖ Project Manager

6406536642079. ✖ Development Lead

Question Number : 13 Question Id : 6406532039946 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A team completes 4 story points in one sprint. This value is used to predict future progress.

What is this value called?

Options :

6406536642080. ✖ Throughput

6406536642081. ✖ Capacity

6406536642082. ✔ Velocity

6406536642083. ✖ Burn rate

Question Number : 14 Question Id : 6406532039947 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Several developers with critical expertise leave the organization midway through a project, causing delays.

Which mitigation strategy BEST addresses this risk?

Options :

6406536642084. ✖ Reducing scope

6406536642085. ✔ Cross-training team members

6406536642086. ✖ Using third-party components

6406536642087. ✖ Increasing automation

Question Number : 15 Question Id : 6406532039948 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A module performs a sequence of operations where each step depends on the previous one, but all steps contribute to one main goal.

This is an example of:

Options :

6406536642088. ✖ Functional cohesion

6406536642089. ✔ Sequential cohesion

6406536642090. ✖ Coincidental cohesion

6406536642091. ✖ Temporal cohesion

Question Number : 16 Question Id : 6406532039949 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If class Book inherits from class Product, which statement is true?

Options :

6406536642092. ✖ Book has-a Product

6406536642093. ✖ Book uses Product

6406536642094. ✔ Book is-a Product

6406536642095. ✖ Product is part of Book

Question Number : 17 Question Id : 6406532039950 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which UML diagram is used to model message exchanges over time among objects?

Options :

- 6406536642096. ✖ Class diagram
- 6406536642097. ✔ Sequence diagram
- 6406536642098. ✖ Activity diagram
- 6406536642099. ✖ Use-case diagram

Question Number : 18 Question Id : 6406532039951 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A developer designs a `Product` class with private data like `price` and `name`, and allows access only through methods such as `getPrice()` and `setPrice()`.

This describes which of the following object-oriented programming concepts?

Options :

- 6406536642100. ✖ Inheritance
- 6406536642101. ✔ Encapsulation
- 6406536642102. ✖ Association
- 6406536642103. ✖ Dependency

Question Number : 19 Question Id : 6406532039952 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A team of 20 developers is working on a large web application. One developer commits faulty code to the main branch that crashes the system. Other developers are unable to test their own changes. The team wants to quickly revert to a stable version and identify who made the faulty change.

Which feature of a Version Control System directly addresses this situation?

Options :

- 6406536642104. ✖ Branching
- 6406536642105. ✔ Version history tracking
- 6406536642106. ✖ Staging area
- 6406536642107. ✖ Distributed repositories

Question Number : 20 Question Id : 6406532039953 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A banking application calculates interest. Due to a wrong formula in the code, the system produces an incorrect interest amount for customers.

In the same banking application, the developer finds that the line

```
interest = balance * 0.5
```

was written instead of

```
interest = balance * 0.05
```

This incorrect line is an example of:

Options :

- 6406536642108. ✖ Failure
- 6406536642109. ✔ Fault
- 6406536642110. ✖ Error
- 6406536642111. ✖ Test case

Question Number : 21 Question Id : 6406532039954 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A video-streaming application is functionally correct, but users complain that it is very slow when playing high-resolution videos.

The development team wants to find out:

- Which functions consume the most CPU time
- Where memory usage is highest
- Which parts of the program are performance bottlenecks

Which debugging tool or technique should the team use?

Options :

- 6406536642112. ✖ Logging
- 6406536642113. ✖ Dump & Diff
- 6406536642114. ✔ Profiling Tool
- 6406536642115. ✖ Backward Debugging

Question Number : 22 Question Id : 6406532039955 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A photo-editing application applies a series of transformations to an image: first resizing, then filtering, then compression, and finally saving it to disk. Each stage takes input from the previous stage and passes output to the next.

Which architecture best represents this system?

Options :

- 6406536642116. ✖ Client-server

6406536642117. ✓ Pipe-and-filter

6406536642118. ✗ Peer-to-peer

6406536642119. ✗ MVC

Question Number : 23 Question Id : 6406532039956 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An e-commerce system performs the following steps when a customer places an order:

1. Create an Order
2. Notify the Seller
3. Call Packaging
4. Call Delivery

Originally, every client had to call all these classes in the correct sequence.

To simplify this, a new class `OrderService` was created with a method:

```
placeOrder(product, seller)
```

This method internally creates all required objects and executes all steps in order.
Which design pattern is being used?

Options :

6406536642120. ✗ Adapter

6406536642121. ✓ Facade

6406536642122. ✗ Builder

6406536642123. ✗ Factory

Question Number : 24 Question Id : 6406532039957 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software company builds a system to process both physical books and e-books.

The interface `OrderService` contains the following methods:

• `placeOrder()`

• `notifySeller()`

• `packItem()`

• `shipItem()`

The e-book module must implement `packItem()` and `shipItem()` even though they do nothing.

Which principle is violated?

Options :

- 6406536642124. ✖ Open-Closed Principle
- 6406536642125. ✔ Interface Segregation Principle
- 6406536642126. ✖ Liskov Substitution Principle
- 6406536642127. ✖ Dependency Inversion Principle

Sub-Section Number :

3

Sub-Section Id :

640653374406

Question Shuffling Allowed :

Yes

Question Number : 25 Question Id : 6406532039937 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are non-functional requirements?
(Select all that apply)

Options :

- 6406536642044. ✔ The system must display new products within 5 seconds
- 6406536642045. ✔ The system should behave consistently over time
- 6406536642046. ✖ The seller can update product prices
- 6406536642047. ✔ The system should recover from invalid input

Question Number : 26 Question Id : 6406532039943 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are characteristics of heuristic (model-based) estimation techniques?
(Select all that apply)

Options :

- 6406536642068. ✔ Use mathematical models
- 6406536642069. ✔ Relate effort to project size
- 6406536642070. ✖ Depend solely on expert intuition
- 6406536642071. ✔ Include models like COCOMO